a **Source-specific response units**

Replogle	Norman	MIX-Seq
K562 · CRISPRi	K562 · CRISPRa	Chemical perturbations
276,108 QC-retained cells	104,507 QC-retained cells	120,003 chemical cells
2,151 gene × GEM units	**2,180 condition × GEM units**	**527 drug × context units**
358 genes · 48 technical GEMs	280 conditions · 8 technical GEMs	13 drugs · 109 cell lines
one published experiment	one published experiment	169 matched assay contexts

Three held-out studies; GEM partitions describe within-experiment sampling.

b **Shared measurements and independent cell samples**

Metadata-defined gene intersection

Replogle · Norman · MIX-Seq

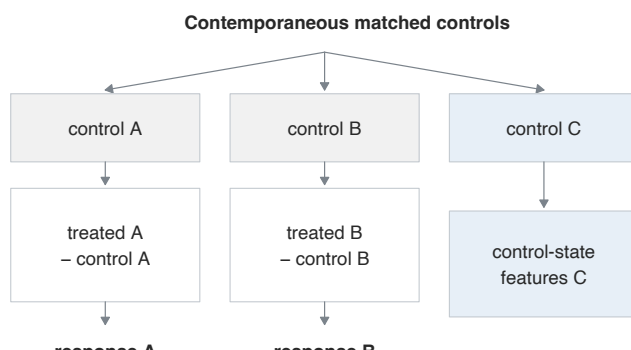
Adamson · Frangieh

7,848 measured genes

Duplicate counts summed before log1p

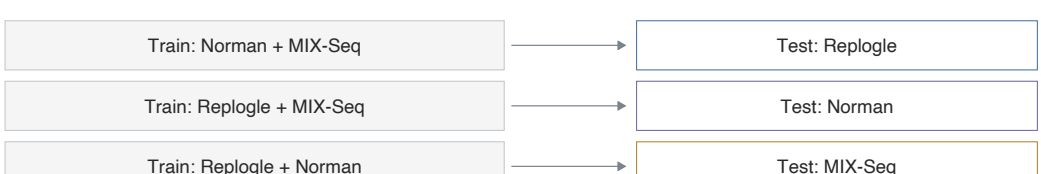
CP10k denominator: the same 7,848 genes

3 signed modules + 9 Hallmark summaries



Disjoint cell subsets

c **Study-held-out prediction**



Scaling, ridge penalties and model fitting use training studies only.